

Heaven's light is our guide

Rajshahi University of Engineering & Technology

Department of Computer Science & Engineering



LAB TASK

Course Code: CSE 3102

Course Name: Database Systems Sessional

Task: Create a Database and some tables. Also, insert a few values and then apply some SQL Queries based on the situation.

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Create the database and the tables:

1. Create a database called “SmartHomeDB”. Write the corresponding query for this in your report.

...

```
CREATE DATABASE SmartHomeDB;  
USE SmartHomeDB;
```

...

2. Create a table titled “Users” with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
user_id	INT	PRIMARY KEY, NOT NULL, AUTO INCREMENT
username	VARCHAR(50)	NOT NULL
email	VARCHAR(100)	NOT NULL
phone	VARCHAR(15)	NOT NULL
address	VARCHAR(200)	NOT NULL

...

```
CREATE TABLE Users (  
    user_id INT PRIMARY KEY NOT NULL AUTO_INCREMENT,  
    username VARCHAR(50) NOT NULL,  
    email VARCHAR(100) NOT NULL,  
    phone VARCHAR(15) NOT NULL,  
    address VARCHAR(200) NOT NULL  
);
```

...

3. Create a table titled “Homes” with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
home_id	INT	PRIMARY KEY, NOT NULL, AUTO INCREMENT
user_id	INT	NOT NULL
home_name	VARCHAR(100)	NOT NULL
address	VARCHAR(200)	NOT NULL

...

```

CREATE TABLE Homes (

    home_id INT PRIMARY KEY NOT NULL AUTO_INCREMENT,

    user_id INT NOT NULL,

    home_name VARCHAR(100) NOT NULL,

    address VARCHAR(200) NOT NULL

);

'''

```

4. Create a table titled “Devices” with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
device_id	INT	PRIMARY KEY, NOT NULL, AUTO INCREMENT
home_id	INT	NOT NULL
device_name	VARCHAR(100)	NOT NULL
device_type	VARCHAR(50)	NOT NULL

```

'''

CREATE TABLE Devices (

    device_id INT PRIMARY KEY NOT NULL AUTO_INCREMENT,

    home_id INT NOT NULL,

    device_name VARCHAR(100) NOT NULL,

    device_type VARCHAR(50) NOT NULL

);

'''

```

5. Create a table titled “Rooms” with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
room_id	INT	PRIMARY KEY, NOT NULL, AUTO INCREMENT
home_id	INT	NOT NULL
room_name	VARCHAR(100)	NOT NULL

```

'''

CREATE TABLE Rooms (

```

```

room_id INT PRIMARY KEY NOT NULL AUTO_INCREMENT,

home_id INT NOT NULL,

room_name VARCHAR(100) NOT NULL

);

'''

```

6. Create a table titled “DeviceStatusLogs” with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
log_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
device_id	INT	NOT NULL
status	VARCHAR(50)	NOT NULL
timestamp	DATETIME	NOT NULL

```

'''

CREATE TABLE DeviceStatusLogs (

    log_id INT PRIMARY KEY NOT NULL AUTO_INCREMENT,

    device_id INT NOT NULL,

    status VARCHAR(50) NOT NULL,

    timestamp DATETIME NOT NULL

);

'''

```

7. Create a table titled “Schedules” with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
schedule_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
device_id	INT	NOT NULL
action	VARCHAR(50)	NOT NULL
scheduled_time	DATETIME	NOT NULL

```

'''

CREATE TABLE Schedules (

    schedule_id INT PRIMARY KEY NOT NULL AUTO_INCREMENT,


```

```

device_id INT NOT NULL,

action VARCHAR(50) NOT NULL,

scheduled_time DATETIME NOT NULL

);
'''

```

8. Create a table titled “Scenes” with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
scene_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
home_id	INT	NOT NULL
scene_name	VARCHAR(100)	NOT NULL
description	TEXT	NULLABLE

```

'''
CREATE TABLE Scenes (

    scene_id INT PRIMARY KEY NOT NULL AUTO_INCREMENT,

    home_id INT NOT NULL,

    scene_name VARCHAR(100) NOT NULL,

    description TEXT

);
'''

```

9. Create a table titled “SceneDevices” with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
scene_device_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
scene_id	INT	NOT NULL
device_id	INT	NOT NULL
desired_status	VARCHAR(50)	NOT NULL

```

'''
CREATE TABLE SceneDevices (

    scene_device_id INT PRIMARY KEY NOT NULL AUTO_INCREMENT,

```

```

scene_id INT NOT NULL,

device_id INT NOT NULL,

desired_status VARCHAR(50) NOT NULL

);

'''

```

10. Create a table titled “Alerts” with the following constraints. Write the corresponding query for this in your report.

Field	Type	Constraints
alert_id	INT	PRIMARY KEY, NOT NULL, AUTO_INCREMENT
home_id	INT	NOT NULL
alert_type	VARCHAR(50)	NOT NULL
alert_message	TEXT	NOT NULL
alert_time	DATETIME	NOT NULL

```

'''

CREATE TABLE Alerts (

    alert_id INT PRIMARY KEY NOT NULL AUTO_INCREMENT,

    home_id INT NOT NULL,

    alert_type VARCHAR(50) NOT NULL,

    alert_message TEXT NOT NULL,

    alert_time DATETIME NOT NULL

);

'''

```

Data Insertion:

11. Populate the table “Users” with the following data. Write the corresponding insertion query for this in your report.

user_id	username	email	phone	address
1	john_doe	john@example.com	01711111111	Dhaka
2	jane_smith	jane@example.com	01722222222	Chittagong
3	rahim	rahim@example.com	01733333333	Rajshahi
4	karim	karim@example.com	01744444444	Sylhet

5	nazia	nazia@example.com	01755555555	Khulna
6	tamim	tamim@example.com	01766666666	Barisal

...

INSERT INTO Users (user_id, username, email, phone, address) VALUES

(1, 'john_doe', 'john@example.com', '01711111111', 'Dhaka'),

(2, 'jane_smith', 'jane@example.com', '01722222222', 'Chittagong'),

(3, 'rahim', 'rahim@example.com', '01733333333', 'Rajshahi'),

(4, 'karim', 'karim@example.com', '01744444444', 'Sylhet'),

(5, 'nazia', 'nazia@example.com', '01755555555', 'Khulna'),

(6, 'tamim', 'tamim@example.com', '01766666666', 'Barisal');

...

12. Populate the table “Homes” with the following data. Write the corresponding insertion query for this in your report.

home_id	user_id	home_name	address
1	1	Doe Villa	123 Dhaka Road
2	2	Smith House	56 Ctg Lane
3	3	Rahim Residence	78 Rajshahi St
4	4	Karim Kutir	91 Sylhet Ave
5	5	Nazia Nibash	34 Khulna Rd
6	6	Tamim Tower	89 Barisal Blvd

...

INSERT INTO Homes (home_id, user_id, home_name, address) VALUES

(1, 1, 'Doe Villa', '123 Dhaka Road'),

(2, 2, 'Smith House', '56 Ctg Lane'),

(3, 3, 'Rahim Residence', '78 Rajshahi St'),

(4, 4, 'Karim Kutir', '91 Sylhet Ave'),

(5, 5, 'Nazia Nibash', '34 Khulna Rd'),

(6, 6, 'Tamim Tower', '89 Barisal Blvd');

'''

13. Populate the table “Devices” with the following data. Write the corresponding insertion query for this in your report.

device_id	home_id	device_name	device_type
1	1	Thermostat	Temperature Control
2	1	Smart Light	Lighting
3	2	Camera	Security
4	3	Smart TV	Entertainment
5	4	Door Lock	Security
6	5	Air Purifier	Health

'''

INSERT INTO Devices (device_id, home_id, device_name, device_type) VALUES

(1, 1, 'Thermostat', 'Temperature Control'),

(2, 1, 'Smart Light', 'Lighting'),

(3, 2, 'Camera', 'Security'),

(4, 3, 'Smart TV', 'Entertainment'),

(5, 4, 'Door Lock', 'Security'),

(6, 5, 'Air Purifier', 'Health');

'''

14. Populate the table “Rooms” with the following data. Write the corresponding insertion query for this in your report.

room_id	home_id	room_name
1	1	Living Room
2	1	Bedroom
3	2	Kitchen
4	3	Bathroom
5	4	Study
6	5	Dining Room

'''

INSERT INTO Rooms (room_id, home_id, room_name) VALUES

(1, 1, 'Living Room'),

(2, 1, 'Bedroom'),


```
(3, 2, 'Kitchen'),
(4, 3, 'Bathroom'),
(5, 4, 'Study'),
(6, 5, 'Dining Room');
'''
```

15. Populate the table “DeviceStatusLogs” with the following data. Write the corresponding insertion query for this in your report.

log_id	device_id	status	timestamp
1	1	ON	2025-06-01 08:00:00
2	2	OFF	2025-06-01 09:00:00
3	3	ON	2025-06-01 10:00:00
4	4	OFF	2025-06-01 11:00:00
5	5	ON	2025-06-01 12:00:00
6	6	OFF	2025-06-01 13:00:00

```
'''
INSERT INTO DeviceStatusLogs (log_id, device_id, status, timestamp) VALUES

(1, 1, 'ON', '2025-06-01 08:00:00'),
(2, 2, 'OFF', '2025-06-01 09:00:00'),
(3, 3, 'ON', '2025-06-01 10:00:00'),
(4, 4, 'OFF', '2025-06-01 11:00:00'),
(5, 5, 'ON', '2025-06-01 12:00:00'),
(6, 6, 'OFF', '2025-06-01 13:00:00');
```

'''

16. Populate the table “Schedules” with the following data. Write the corresponding insertion query for this in your report.

schedule_id	device_id	action	scheduled_time
1	1	Turn On	2025-06-02 08:00:00
2	2	Turn Off	2025-06-02 09:00:00
3	3	Turn On	2025-06-02 10:00:00
4	4	Turn Off	2025-06-02 11:00:00
5	5	Turn On	2025-06-02 12:00:00

6	6	Turn Off	2025-06-02 13:00:00
---	---	----------	---------------------

...

INSERT INTO Schedules (schedule_id, device_id, action, scheduled_time) VALUES

(1, 1, 'Turn On', '2025-06-02 08:00:00'),
 (2, 2, 'Turn Off', '2025-06-02 09:00:00'),
 (3, 3, 'Turn On', '2025-06-02 10:00:00'),
 (4, 4, 'Turn Off', '2025-06-02 11:00:00'),
 (5, 5, 'Turn On', '2025-06-02 12:00:00'),
 (6, 6, 'Turn Off', '2025-06-02 13:00:00');

...

17. Populate the table “Scenes” with the following data. Write the corresponding insertion query for this in your report.

scene_id	home_id	scene_name	description
1	1	Morning Routine	Turn on lights, start coffee maker
2	2	Night Mode	Dim lights, lock doors
3	3	Away Mode	Activate security systems
4	4	Study Mode	Bright light, mute TV
5	5	Dinner Setup	Dim lights, play music
6	6	Relax	Play soft music, aroma diffuser on

...

INSERT INTO Scenes (scene_id, home_id, scene_name, description) VALUES

(1, 1, 'Morning Routine', 'Turn on lights, start coffee maker'),
 (2, 2, 'Night Mode', 'Dim lights, lock doors'),
 (3, 3, 'Away Mode', 'Activate security systems'),
 (4, 4, 'Study Mode', 'Bright light, mute TV'),
 (5, 5, 'Dinner Setup', 'Dim lights, play music'),
 (6, 6, 'Relax', 'Play soft music, aroma diffuser on');

...

18. Populate the table “SceneDevices” with the following data. Write the corresponding insertion query for this in your report.

scene_device_id	scene_id	device_id	desired_status
1	1	1	ON
2	1	2	ON
3	2	3	ON
4	3	4	OFF
5	4	5	ON
6	5	6	ON

```

```
INSERT INTO SceneDevices (scene_device_id, scene_id, device_id, desired_status)
VALUES
```

```
(1, 1, 1, 'ON'),
```

```
(2, 1, 2, 'ON'),
```

```
(3, 2, 3, 'ON'),
```

```
(4, 3, 4, 'OFF'),
```

```
(5, 4, 5, 'ON'),
```

```
(6, 5, 6, 'ON');
```

```

19. Populate the table “Alerts” with the following data. Write the corresponding insertion query for this in your report.

alert_id	home_id	alert_type	alert_message	alert_time
1	1	Fire	Smoke detected in kitchen	2025-06-03 14:00:00
2	2	Intrusion	Motion detected at front door	2025-06-03 15:00:00
3	3	Gas Leak	Gas leak in kitchen	2025-06-03 16:00:00
4	4	Temperature	High temperature detected	2025-06-03 17:00:00
5	5	Humidity	High humidity detected	2025-06-03 18:00:00
6	6	Power	Power outage	2025-06-03 19:00:00

```

```
INSERT INTO Alerts (alert_id, home_id, alert_type, alert_message, alert_time) VALUES
```

```
(1, 1, 'Fire', 'Smoke detected in kitchen', '2025-06-03 14:00:00'),
```

```
(2, 2, 'Intrusion', 'Motion detected at front door', '2025-06-03 15:00:00'),
```

```
(3, 3, 'Gas Leak', 'Gas leak in kitchen', '2025-06-03 16:00:00'),
(4, 4, 'Temperature', 'High temperature detected', '2025-06-03 17:00:00'),
(5, 5, 'Humidity', 'High humidity detected', '2025-06-03 18:00:00'),
(6, 6, 'Power', 'Power outage', '2025-06-03 19:00:00');
'''
```

## SQL Queries to do the following tasks:

**a) Query 1:** Show all users and their email addresses.

```
'''
```

```
SELECT username, email FROM Users;
```

```
'''
```

| username   | email             |
|------------|-------------------|
| john_doe   | john@example.com  |
| jane_smith | jane@example.com  |
| rahim      | rahim@example.com |
| karim      | karim@example.com |
| nazia      | nazia@example.com |
| tamim      | tamim@example.com |

**b) Query 2:** List all devices with their names and types.

```
'''
```

```
SELECT device_name, device_type FROM Devices;
```

```
'''
```

| device_name  | device_type         |
|--------------|---------------------|
| Thermostat   | Temperature Control |
| Smart Light  | Lighting            |
| Camera       | Security            |
| Smart TV     | Entertainment       |
| Door Lock    | Security            |
| Air Purifier | Health              |

**c) Query 3:** Show all homes along with the username of the owner.

'''

```
SELECT H.home_name, U.username
FROM Homes H
JOIN Users U ON H.user_id = U.user_id;
```

'''

| home_name       | username   |
|-----------------|------------|
| Doe Villa       | john_doe   |
| Smith House     | jane_smith |
| Rahim Residence | rahim      |
| Karim Kutir     | karim      |
| Nazia Nibash    | nazia      |
| Tamim Tower     | tamim      |

**d) Query 4:** List all devices that are currently ON.

'''

```
SELECT D.device_name, L.status, L.timestamp
FROM Devices D
JOIN DeviceStatusLogs L ON D.device_id = L.device_id
WHERE L.status = 'ON';
```

'''

| device_name | status | timestamp           |
|-------------|--------|---------------------|
| Thermostat  | ON     | 2025-06-01 08:00:00 |
| Camera      | ON     | 2025-06-01 10:00:00 |
| Door Lock   | ON     | 2025-06-01 12:00:00 |

e) **Query 5:** Find all devices scheduled to Turn Off after 10 AM on 2025-06-02.

'''

```
SELECT D.device_name, S.action, S.scheduled_time
FROM Devices D
JOIN Schedules S ON D.device_id = S.device_id
WHERE S.action = 'Turn Off' AND S.scheduled_time > '2025-06-02 10:00:00';
```

'''

| device_name  | action   | scheduled_time      |
|--------------|----------|---------------------|
| Smart TV     | Turn Off | 2025-06-02 11:00:00 |
| Air Purifier | Turn Off | 2025-06-02 13:00:00 |

f) **Query 6:** List all rooms in “Doe Villa”.

'''

```
SELECT H.home_name, R.room_name
FROM Rooms R
JOIN Homes H ON R.home_id = H.home_id
WHERE H.home_name = 'Doe Villa';
```

'''

| home_name | room_name   |
|-----------|-------------|
| Doe Villa | Living Room |
| Doe Villa | Bedroom     |

**g) Query 7:** Get all scenes for the home owned by 'karim'.

'''

```
SELECT S.scene_name, S.description
```

```
FROM Scenes S
```

```
JOIN Homes H ON S.home_id = H.home_id
```

```
JOIN Users U ON H.user_id = U.user_id
```

```
WHERE U.username = 'karim';
```

'''

| username | scene_name | description           |
|----------|------------|-----------------------|
| karim    | Study Mode | Bright light, mute TV |

**h) Query 8:** Count how many alerts were generated for each alert type.

'''

```
SELECT alert_type, COUNT(*) AS total_alerts
```

```
FROM Alerts
```

```
GROUP BY alert_type;
```

'''

| alert_type  | total_alerts |
|-------------|--------------|
| Fire        | 1            |
| Gas Leak    | 1            |
| Humidity    | 1            |
| Intrusion   | 1            |
| Power       | 1            |
| Temperature | 1            |

**i) Query 9:** Show device names that are part of more than one scene.

'''

```

SELECT D.device_name, COUNT(SD.scene_id) AS scene_count
FROM SceneDevices SD
JOIN Devices D ON SD.device_id = D.device_id
GROUP BY SD.device_id
HAVING COUNT(SD.scene_id) > 1;
'''

```

| device_name | scene_count |
|-------------|-------------|
|-------------|-------------|

**j) Query 10:** For each home, list the number of devices, rooms, and alerts.

```

'''
SELECT H.home_name,
 COUNT(DISTINCT D.device_id) AS device_count,
 COUNT(DISTINCT R.room_id) AS room_count,
 COUNT(DISTINCT A.alert_id) AS alert_count
FROM Homes H
LEFT JOIN Devices D ON H.home_id = D.home_id
LEFT JOIN Rooms R ON H.home_id = R.home_id
LEFT JOIN Alerts A ON H.home_id = A.home_id
GROUP BY H.home_id;
'''

```

| home_name       | device_count | room_count | alert_count |
|-----------------|--------------|------------|-------------|
| Doe Villa       | 2            | 2          | 1           |
| Smith House     | 1            | 1          | 1           |
| Rahim Residence | 1            | 1          | 1           |
| Karim Kutir     | 1            | 1          | 1           |
| Nazia Nibash    | 1            | 1          | 1           |
| Tamim Tower     | 0            | 0          | 1           |